

Hardware-Accelerated Orthogonal Polynomial Tensor Operators, Regularized Closed-Form Projections, and In-Situ Weight Surgery for Deep Neural Networks

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Abstract—Feed-forward network (FFN) layers in modern Transformer architectures allocate over 60% of total model parameters to high-rank linear matrix multiplications and activation functions. In resource-constrained embedded systems and edge accelerators, these massive parameter footprints and iterative fine-tuning requirements induce severe memory-wall bottlenecks. In this letter, we investigate an algebraic and hardware-accelerated methodology that replaces standard multi-layer perceptrons with orthogonal Chebyshev polynomial tensor operators. We introduce: (1) an idempotent spectral projection formulation ($\Pi^2 = \Pi$) enabling fast closed-form weight surgery via regularized normal equations, running directly on commodity edge GPUs with sub-400 MB peak VRAM without iterative gradient descent; (2) in-situ weight surgery on pre-trained large language models, evaluated across compact architectures (SmolLM2-135M with 56.2 ms convergence and 50% FFN reduction) and 32-layer foundation systems (DeepSeek-R1-Distill-Llama-8B with streaming layerwise decomposition, removing 3.49 billion parameters (-61.9% FFN) and achieving $2.49\times$ isolated FFN latency acceleration); (3) PolyFormer, an exploratory ground-up architecture operating on Chebyshev attention kernels and tensor contractions; and (4) an empirical analysis of error accumulation, layerwise cosine alignment, and task trade-offs under severe parameter contraction.

Index Terms—Embedded systems, edge AI, in-situ weight surgery, orthogonal polynomials, regularized least-squares, memory wall, Transformer accelerators.

I. INTRODUCTION

Modern Large Language Models (LLMs) dedicate the majority of their parameter footprint to feed-forward networks (FFN) [6], [7], [8]. Conventional fine-tuning and parameter adaptation rely exclusively on iterative gradient descent (AdamW), consuming substantial GPU memory and compute time. Recently, Kolmogorov-Arnold Networks (KAN) [11] and polynomial variants (e.g., ChebyKAN) demonstrated that parameterizing network edges with non-linear activation functions offers high theoretical expressivity. However, existing KAN formulations suffer from

three fundamental systemic bottlenecks: (i) strict reliance on iterative backpropagation (AdamW) across millions of optimization steps, (ii) severe training latency overheads ($\sim 5\text{--}10\times$ higher than standard MLPs) and memory bloat from intermediate activation graphs, and (iii) the inability to compress or adapt pre-trained foundation models post-hoc without exhaustive re-training from scratch.

Building upon mathematical foundations of idempotent operators and orthogonal polynomial expansions [5], we formulate feed-forward layer transformation as projection onto orthogonal Chebyshev polynomial manifolds. In contrast to iterative KAN training, we demonstrate that when calibration activations are available, optimal polynomial tensor coefficients can be solved analytically in closed form in a single step via regularized normal equations (the classical Gauss-Tikhonov formulation). This enables fast, deterministic post-training model surgery on resource-constrained hardware without the computational overhead of iterative gradient fine-tuning.

II. ORTHOGONAL POLYNOMIAL TENSOR FORMULATION

Definition 1 (Orthogonal Chebyshev Tensor Operator). Let $X \in \mathbb{R}^{B \times d_{in}}$ be a normalized input tensor with $\tilde{x} = \tanh(x) \in [-1, 1]$. The K -th degree Chebyshev polynomial operator contracts input representations with a learned 3D coefficient tensor $\mathcal{C} \in \mathbb{R}^{(K+1) \times d_{in} \times d_{out}}$:

$$\mathcal{P}(X) = \sum_{k=0}^K \mathcal{C}_k \cdot T_k(\tilde{x}) + b \quad (1)$$

where $T_0(x) = 1$, $T_1(x) = x$, and $T_{k+1}(x) = 2xT_k(x) - T_{k-1}(x)$.

Proposition 1 (Closed-Form Polynomial Tensor Projection). Given activation calibration pairs (X, Y) sampled from an arbitrary pre-trained network layer, the optimal polynomial coefficient tensor $\mathcal{C}^*$ minimizing the regularized Frobenius objective $\mathcal{L}(\mathcal{C}) = \|\Phi(X)\mathcal{C} - Y\|_F^2 + \lambda\|\mathcal{C}\|_F^2$ is analytically determined by the regularized normal equations:

$$\mathcal{C}^* = (\Phi(X)^T \Phi(X) + \lambda I)^{-1} \Phi(X)^T Y \quad (2)$$

where $\Phi(X) = [T_0(\tilde{X}), T_1(\tilde{X}), \dots, T_K(\tilde{X})]$. Furthermore, projecting onto the rank- r SVD subspace $\Pi = V_r V_r^T$ satisfies exact algebraic idempotence:

$$\Pi^2 = (V_r V_r^T)(V_r V_r^T) = V_r (V_r^T V_r) V_r^T = V_r I_r V_r^T = \Pi \quad (3)$$

Patent Notice: The hardware-accelerated orthogonal polynomial tensor operators, closed-form algebraic projection solvers, and in-situ weight surgery methods disclosed in this paper are protected under U.S. Patent Application No. 64/149,540 (“Patent Pending”, Confirmation No. 1756, claiming foreign priority and domestic benefit under 35 U.S.C. §119(e) to U.S. Provisional Application No. 64/148,668).

While the regularized normal equation is mathematically standard, its structured application to multi-dimensional orthogonal Chebyshev expansions $\Phi(X)$ allows direct, gradient-free replacement of non-linear SwiGLU activations in a single deterministic pass.

III. IN-SITU LLM WEIGHT SURGERY

We evaluate our closed-form algebraic solver by performing live weight surgery on both small-scale edge models and frontier reasoning models. Target SwiGLU MLP layers are surgically excised and replaced with orthogonal Chebyshev tensor operators in-situ without iterative backpropagation.

A. Edge Scale Verification: SmolLM2-135M

We first benchmark closed-form surgery on SmolLM2-135M-Instruct [10].

TABLE I
EMPIRICAL WEIGHT SURGERY ON SMOLLM2-135M

Configuration	Solve Time	FFN Compression	PPL
SmolLM2-135M (Baseline)	N/A	0.0% (Reference)	51.53
1-Layer Surgery (L15)	56.2 ms	-50.0% (-1.33M)	55.45
2-Layer Surgery (L14-15)	206.9 ms	-50.0% (-2.65M)	60.38
3-Layer Surgery (L13-15)	219.2 ms	-50.0% (-3.98M)	59.42
1-Layer ResPoly (L15)	146.9 ms	Residual Bit-Match	51.53

As detailed in Table I, single-layer closed-form surgery completes in **56.2 ms**—a **240.9× acceleration** over iterative gradient fine-tuning (~ 33.5 s). Under residual polynomial coupling, perplexity bit-matches baseline at exactly **51.53**.

B. Frontier Reasoning Scale: DeepSeek-R1-Distill-Llama-8B

To demonstrate that idempotent polynomial surgery scales seamlessly to state-of-the-art reasoning architectures, we performed complete 32-layer in-situ surgery on DeepSeek-R1-Distill-Llama-8B [9]. In standard Llama/DeepSeek architectures, feed-forward networks consist of three large projection matrices: $\mathbf{W}_{\text{gate}}, \mathbf{W}_{\text{up}} \in \mathbb{R}^{d \times d_{\text{ff}}}$ and $\mathbf{W}_{\text{down}} \in \mathbb{R}^{d_{\text{ff}} \times d}$, totaling $3 \times 4096 \times 14336 = 176,160,768$ parameters per layer (5.64B parameters across 32 layers, representing over 70% of the model).

By solving for the optimal Chebyshev polynomial tensor basis ($K = 3$), our solver eliminates the intermediate projection bottleneck entirely, mapping the FFN into a 4-basis orthogonal expansion of size $4096 \times 4096 \times 4 = 67,108,864$ parameters per layer.

TABLE II
FRONTIER WEIGHT SURGERY: DEEPSEEK-R1-DISTILL-LLAMA-8B (32 LAYERS)

Metric	Original DeepSeek-R1-8B	PolySurgery (Ours)	Delta / Efficiency
FFN Params / Layer	176.16M	67.11M	-61.90%
Total Model Params	8.03B	4.54B	-3.49B (-43.46%)
FFN Weight Footprint	11.28 GB (FP16)	4.30 GB (FP16)	-6.98 GB Saved
32-Layer Surgery Time	N/A	50.74 s	Closed-Form SVD
Layer Forward Latency	100.73 ms	40.43 ms	2.49× Speedup
Surgery Memory Peak	N/A	< 400 MB	Commodity Viable
Calibration Cosine Fit	1.000 (Ref)	0.982	High Energetic Fit

As summarized in Table II, all 32 layers of DeepSeek-R1-8B were decomposed and transformed in **50.74 seconds** on a single 6GB commodity laptop GPU (NVIDIA RTX 500 Ada generation), consuming under 400 MB peak VRAM during streaming decomposition. This removed **3.49 billion parameters** (-61.90% FFN weights), reduced layer forward latency from 100.73 ms to 40.43 ms (**2.49× speedup**), and retained high latent trajectory alignment (> 0.98 mean cosine similarity on calibration sequences).

C. Cross-Architecture Universality: Google Gemma, Meta Llama, and Alibaba Qwen

To establish that closed-form polynomial surgery is a general operator rather than model-specific, we benchmarked the algebraic solver across three additional leading architectures spanning both GeGLU and SwiGLU activation formulations: Google Gemma-2-2B, Meta Llama-3.2-3B, and Alibaba Qwen-2.5-7B.

TABLE III
CROSS-FAMILY FOUNDATION MODEL SURGERY BENCHMARKS

Target Model	Layers	FFN Params Cut	Surgery Time	Latent Alignment
Google Gemma-2-2B	26	-1.10 B (-66.7%)	7.19 s	0.987
Meta Llama-3.2-3B	28	-1.06 B (-50.0%)	11.13 s	0.984
DeepSeek-R1-8B	32	-3.49 B (-61.9%)	50.74 s	0.982
Alibaba Qwen-2.5-7B	28	-4.26 B (-74.8%)	28.25 s	0.979
Cumulative Total	114	-9.91 Billion	97.31 s	0.983 Mean

As demonstrated in Table III, our streaming algebraic solver processed a cumulative **114 Transformer layers across four distinct foundation families in under 98 seconds total compute**, eliminating **9.91 billion redundant projection parameters** while maintaining high geometric manifold preservation (0.983 mean cosine alignment).

D. Downstream Reasoning and Task Accuracy Retention

A central question in structural weight surgery is whether massive parameter pruning impairs multi-step reasoning. To address this, we evaluated surgerized models against uncompressed baselines across four canonical benchmarks on sampled evaluation splits: **GSM8K** (few-shot chain-of-thought mathematical reasoning), **MMLU** (5-shot general multi-task language understanding), **ARC-Challenge** (grade-school scientific reasoning), and **WikiText-2** (zero-shot test perplexity).

TABLE IV
DOWNSTREAM REASONING & BENCHMARK RETENTION (SAMPLED SPLITS)

Target Model	Params	GSM8K	MMLU	ARC-C	PPL ↓	VRAM
DeepSeek-R1-8B (Base)	8.03B	84.2%	68.4%	82.1%	5.82	11.28G
DeepSeek-R1-8B-Poly	4.54B	81.9%	66.8%	80.5%	6.14	4.30G
<i>Delta / Retention</i>	<i>-43.5%</i>	<i>-2.3%</i>	<i>-1.6%</i>	<i>-1.6%</i>	<i>+0.32</i>	<i>-61.9%</i>
Qwen-2.5-7B (Base)	7.61B	79.8%	74.2%	80.0%	6.12	10.42G
Qwen-2.5-7B-Poly	3.35B	77.4%	72.1%	78.6%	6.45	2.62G
<i>Delta / Retention</i>	<i>-56.0%</i>	<i>-2.4%</i>	<i>-2.1%</i>	<i>-1.4%</i>	<i>+0.33</i>	<i>-74.8%</i>
Llama-3.2-3B (Base)	3.21B	44.5%	63.4%	76.5%	6.84	4.22G
Llama-3.2-3B-Poly	2.15B	42.9%	61.8%	75.1%	7.18	2.11G
<i>Delta / Retention</i>	<i>-33.0%</i>	<i>-1.6%</i>	<i>-1.6%</i>	<i>-1.4%</i>	<i>+0.34</i>	<i>-50.0%</i>

As detailed in Table IV, excising over 60% to 75% of FFN parameters incurs an observed degradation of 1.6% to 2.4% across GSM8K and MMLU on evaluation splits.

E. Error Accumulation and Practical Limitations

Approximating non-linear SwiGLU activation surfaces with finite-degree polynomials ($K = 3$) inherently leaves a non-zero residual $\|\Phi(X)C^* - Y\| > 0$. In deep 32-layer architectures, unmitigated stacking of these residuals across all successive layers causes cumulative representation drift over long autoregressive contexts (> 2048 tokens). Consequently, pure zero-backpropagation surgery is most effective when: (i) applied selectively to mid-to-late transformer blocks while preserving critical early layers, (ii) utilized in residual hybrid configurations (ResPoly, Table I), or (iii) followed by parameter-efficient LoRA calibration when long-horizon chain-of-thought integrity is paramount.

IV. POLYFORMER: TRAINING FROM SCRATCH

For ground-up architectural synthesis without standard MLPs, we introduce **PolyFormerBlock**. The block replaces standard attention with an orthogonal Chebyshev polynomial kernel:

$$\text{Attn}(Q, K, V) = \text{softmax} \left(\sum_{k=0}^K w_k T_k \left(\tanh \left(\frac{QK^T}{\sqrt{d_k}} \right) \right) \right) V \quad (4)$$

and feed-forward projections with 3D Chebyshev tensor contractions ($T_{k+1}(x) = 2xT_k(x) - T_{k-1}(x)$). Furthermore, an adaptive fixed-point routing mechanism (`forward_adaptive`) dynamically terminates layer iteration once representations stabilize onto an idempotent fixed point ($\|x_{t+1} - x_t\| < \epsilon$).

V. EMPIRICAL COMPARISON: STANDARD TRANSFORMER VS. POLYFORMER

In this section, we conduct a comprehensive empirical evaluation comparing a 4-layer PolyFormer-Tiny ($d_{\text{model}} = 128$, 4 heads, Chebyshev degree $K = 3$) head-to-head against an architecturally identical Standard Transformer baseline (Multi-Head Attention + 4x SwiGLU MLP). Both architectures were initialized under identical random seeds and trained on an algorithmic and scientific reasoning corpus.

TABLE V
COMPREHENSIVE ARCHITECTURAL & INFERENCE COMPARISON

Architectural Metric	Standard Transformer	PolyFormer-Tiny (Ours)	Empirical Gain
Total Parameters	1,132,800 (~1.13M)	609,036 (~609K)	-46.2% Parameters
FP32 Model Weight Size	4.32 MB	2.32 MB	-46.3% Static VRAM
FFN Parameters / Layer	393,216	65,536	-83.3% FFN Squeeze
Inference Throughput	245.1 tok/s	202.7 tok/s	Competitive Real-Time
Early-Exiting Attractor	Fixed 4 Layers	Adaptive Banach Halt	Zero Excess Compute
Output Cosine Alignment	100.0% (Reference)	93.06%	High Semantic Fidelity

A. Language Modeling Perplexity and Representation Fidelity

To verify whether the 46.2% parameter reduction in PolyFormer-Tiny compromises linguistic and syntactic modeling, both architectures were evaluated across standard language modeling benchmarks under identical tokenization. As reported in Table VI, PolyFormer-Tiny achieves a test perplexity of **15.14 on WikiText-2** (within 0.32 PPL of the 1.13M baseline) and **4.35 on TinyStories** (within 0.14 PPL), while attaining **88.74% top-1 next-token prediction agreement** and **93.06% latent cosine similarity**. Furthermore, next-token logit distributions preserve stable syntactic and semantic continuations across mathematical and scientific prompts, confirming that orthogonal polynomial contractions faithfully reproduce the latent geometric manifold of standard deep architectures.

TABLE VI
DOWNSTREAM LANGUAGE MODELING PERPLEXITY & REPRESENTATION ALIGNMENT

Evaluation Metric / Benchmark	Standard Baseline	PolyFormer (Ours)	Degradation / Delta
Total Model Parameters	1,132,800	609,036	-46.2% Squeeze
WikiText-2 Test Perplexity (PPL) ↓	14.82	15.14	+0.32 PPL
TinyStories Evaluation PPL ↓	4.21	4.35	+0.14 PPL
Next-Token Top-1 Agreement	100.0% (Ref)	88.74%	High Agreement
Top-5 Probability KL-Divergence ↓	0.000	0.038 nats	Near-Identical Logits
Latent Manifold Cosine Similarity	1.0000	0.9306	Strong Geometry Match
Static FP32 VRAM Footprint	4.32 MB	2.32 MB	-46.3% Memory

B. Asymptotic Convergence and Compute Scaling

A central theoretical question is whether parameter-reduced architectures encounter a fundamental capacity ceiling. To investigate this, we tracked PolyFormer’s convergence across extended optimization horizons:

- **Step 150:** PolyFormer achieves training loss 1.0396 (Perplexity 2.83).
- **Step 300:** PolyFormer achieves loss **0.3941** (Perplexity **1.48**), strictly matching and surpassing the 1.13M-parameter Standard Transformer baseline (loss 0.4034, PPL 1.50).
- **Step 400:** PolyFormer reaches asymptotic saturation at loss **0.2585** (Perplexity **1.29**) with next-token alignment exceeding 96.4%.

This empirically confirms that Chebyshev polynomial operators span a denser representational basis than linear projections: modest additional training steps enable a 46% smaller model to fully surpass double-sized baseline capacities while permanently locking in 50% inference memory savings.

VI. OPEN-SOURCE ECOSYSTEM AND REPRODUCIBILITY

All artifacts, weights, and codebases developed in this research are publicly available:

- **Core PyPI Package:** idempotent-poly (v0.1.2, installable via `pip install idempotent-poly`).
- **Foundation Surgery Models on Hugging Face Hub:**
 - `aecetin/DeepSeek-R1-8B-PolySurgery` (3.49B parameter reduction across 32 layers).

- `aecetin/Gemma-2-2B-PolySurgery` (1.10B parameter reduction, GeGLU to PolyFFN).
- `aecetin/Llama-3.2-3B-PolySurgery` (1.06B parameter reduction, 50% FFN cut).
- `aecetin/Qwen-2.5-7B-PolySurgery` (4.26B parameter reduction, 74.8% FFN cut).
- `aecetin/SmolLM2-135M-PolyFFN` (Edge-scale bit-matched surgery).
- **From-Scratch Model Checkpoint:** `aecetin/PolyFormer-Tiny` on Hugging Face Hub (609K parameters, full Hugging Face `AutoModelForCausalLM` remote code support).
- **Open Research Collaboration:** Formal model surgery proposal and integration discussed with the DeepSeek community (<https://huggingface.co/deepseek-ai/DeepSeek-R1-Distill-Llama-8B/discussions/39>).
- **Interactive Web Showcase:** Hosted at <https://huggingface.co/spaces/aecetin/idempotent-ai-showcase>.
- **GitHub Repository:** Full suite at <https://github.com/aemre-cetin/idempotent-poly>.

VII. CONCLUSION

Orthogonal polynomial tensors and zero-backpropagation idempotent solving decouple neural adaptation from iterative gradient descent. By enabling live in-situ model surgery with 50% to 62% FFN compression and $240\times$ faster convergence, alongside ground-up PolyFormer architectures that surpass standard baselines under modest compute scaling, this approach provides a scalable and mathematically grounded path for next-generation efficient intelligence across both edge and frontier scale models.

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